WORKED SOLUTION

STEP I 2016, Question 12

Probability and Statistics • Binomial Symmetry

STEP I 2016, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s1-q12

This question has a full guided treatment in *Beyond STEP*, Section 12.6, *One More Coin*: https://beyondstep.keithmcnulty.org/ch12/one-more-coin.html

The key idea

Parts (i) and (ii) are direct calculations with binomial probabilities, and both give the same strikingly simple answer. Part (iii) explains why. Split Bob's tosses into the first n and one extra toss: for the first n tosses Bob and Alice are on an equal footing, so symmetry does the work, and the extra toss breaks ties fairly.

Solution

Throughout, let A be the number of heads Alice gets and B the number Bob gets. With m tosses of a fair coin, the number of heads X has

$$\mathrm{P}(X = k) = \binom{m}{k}\frac{1}{2^m}, \qquad k = 0, 1, \ldots, m,$$

and Alice's and Bob's tosses are independent, so we can multiply probabilities for A and B.

Part (i)

Alice tosses twice and Bob three times, so

k	0	1	2	3
$\mathrm{P}(A = k)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$	
$\mathrm{P}(B = k)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

Condition on Alice's score. Given $A = a$, we need $B > a$, so we need the tail probabilities $\mathrm{P}(B > 0) = \frac{7}{8}$, $\mathrm{P}(B > 1) = \frac{1}{2}$, $\mathrm{P}(B > 2) = \frac{1}{8}$. Hence

$$\begin{aligned}\mathrm{P}(B > A) &= \tfrac{1}{4} \cdot \tfrac{7}{8} + \tfrac{1}{2} \cdot \tfrac{1}{2} + \tfrac{1}{4} \cdot \tfrac{1}{8} \\ &= \frac{7+8+1}{32} = \frac{1}{2}.\end{aligned}$$

Part (ii)

Now $\mathrm{P}(A = a) = \frac{1}{8}, \frac{3}{8}, \frac{3}{8}, \frac{1}{8}$ for $a = 0, 1, 2, 3$, and $\mathrm{P}(B = k) = \frac{1}{16}, \frac{4}{16}, \frac{6}{16}, \frac{4}{16}, \frac{1}{16}$ for $k = 0, \ldots, 4$. The tail probabilities are

$$\mathrm{P}(B > 0) = \tfrac{15}{16}, \quad \mathrm{P}(B > 1) = \tfrac{11}{16}, \quad \mathrm{P}(B > 2) = \tfrac{5}{16}, \quad \mathrm{P}(B > 3) = \tfrac{1}{16}.$$

So

$$\begin{aligned}\mathrm{P}(B > A) &= \tfrac{1}{8} \cdot \tfrac{15}{16} + \tfrac{3}{8} \cdot \tfrac{11}{16} + \tfrac{3}{8} \cdot \tfrac{5}{16} + \tfrac{1}{8} \cdot \tfrac{1}{16} \\ &= \frac{15+33+15+1}{128} = \frac{64}{128} = \frac{1}{2}.\end{aligned}$$

Once again the answer is exactly $\frac{1}{2}$.

Part (iii)

First, a fact about the equal-tosses game. When Alice and Bob each toss n times, exactly one of "Bob more", "same", "Alice more" happens. The situation is symmetric between the two players (they do identical experiments), so "Alice more" also has probability p_2. Therefore

$$2p_2 + p_1 = 1. \qquad (*)$$

Now let Alice toss n times and Bob $n+1$ times. Let B' be the number of heads in Bob's *first* n tosses, and let H be the event that his last toss is a head. Then $B = B' + 1$ if H happens and $B = B'$ otherwise.

Ask yourself How does comparing B with A depend on comparing B' with A?

Since B' and A are whole numbers and B is either B' or $B' + 1$:

- if $B' > A$, then certainly $B > A$, whatever the last toss;
- if $B' = A$, then $B > A$ exactly when the last toss is a head;
- if $B' < A$, then $B' \leq A - 1$, so $B \leq A$ and Bob does not win.

The first n tosses of Bob against Alice's n tosses is exactly the equal-tosses game, so $\mathrm{P}(B' > A) = p_2$ and $\mathrm{P}(B' = A) = p_1$. The last toss is independent of everything else and $\mathrm{P}(H) = \frac{1}{2}$. Hence

$$\mathrm{P}(\text{Bob gets more heads than Alice}) = p_2 + \tfrac{1}{2}p_1.$$

By $(*)$, $p_2 + \frac{1}{2}p_1 = \frac{1}{2}(2p_2 + p_1) = \frac{1}{2}$.

This is the generalisation: *if Alice tosses a fair coin n times and Bob tosses a fair coin $n+1$ times, the probability that Bob gets more heads than Alice is $\frac{1}{2}$, for every $n \geq 0$.* Parts (i) and (ii) are the cases $n = 2$ and $n = 3$.

Checks

- $n = 0$: Alice tosses nothing and Bob tosses once; Bob wins exactly when he gets a head, with probability $\frac{1}{2}$.
- $n = 2$: by hand, $p_1 = \frac{1}{16} + \frac{4}{16} + \frac{1}{16} = \frac{3}{8}$, and from $(*)$ $p_2 = \frac{5}{16}$. Then $p_2 + \frac{1}{2}p_1 = \frac{5}{16} + \frac{3}{16} = \frac{1}{2}$, agreeing with part (i).
- Exact enumeration by computer confirms $\mathrm{P}(B > A) = \frac{1}{2}$ for every n up to 29, and a simulation of part (ii) with 200 000 trials gave 0.4997.

Common slips

- Assuming $p_2 = \frac{1}{2}$ in the equal-tosses game: ties have positive probability, so $p_2 = \frac{1}{2}(1 - p_1) < \frac{1}{2}$.
- Forgetting the case $B' = A$ in part (iii), or counting it fully rather than with the factor $\frac{1}{2}$ for the extra toss.
- Using tail probabilities $\mathrm{P}(B \geq a)$ instead of $\mathrm{P}(B > a)$.

Beyond STEP

The neat answer comes from a symmetry argument plus conditioning on one extra step, a pattern that runs right through probability, from random walks to ballot problems. A slicker version: because Bob has one more toss, he *fails* to get more heads than Alice exactly when he gets more tails than her. Swapping heads and tails shows "more heads" and "more tails" have equal probability, and they are complementary, so each has probability $\frac{1}{2}$. Grimmett and Welsh, *Probability: An Introduction*, develops these conditioning and symmetry techniques.

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